

John D. Duncan, III

Phone (240) 688-7187
Location Washington, DC
Status US Citizen

Email cv@johnduncan.io
Homepage <https://johnduncan.io>
Github <https://github.com/johndduncaniiii>
Linkedin <https://linkedin.com/in/johndduncan>

MOTIVATION

Making things that are useful to people and that work well.

PROFESSIONAL PROJECTS

Quorum Analytics

Washington, DC

As an early engineer at Quorum Analytics, the best-in-class public affairs civic tech platform, I drove growth from \$9M to \$64M+ ARR as a technical lead by consistently solving Quorum's most complex challenges to deliver over a dozen projects on time and under budget. During my tenure, I authored 5k+ commits, reviewed 1k+ pull requests, and mentored dozens of engineers. Most recently, I launched PAC Hybrid, expanding Quorum's business into the \$15M+ PAC market:

1. Discovered critical discrepancies in Federal Election Commission (FEC) national party committee C-number ids across advisory opinions, website data, and official GitHub repository spanning 50+ years of legal records.
2. Developed end-to-end search, filtering, and data visualization features for four datasets, including backend sql query methods, API design, and frontend UI components.
3. Designed a database synchronization infrastructure to instantly sync billions of rows between Quorum and legacy CisionPAC databases.
4. Reduced database query latency from 60 seconds to 20 milliseconds by optimizing Postgres partition and schema constraints, improving transaction dataset query performance by over 3000x and permanently fixing a 10+ year old bug.
5. Coached engineers to optimize a database migration by replacing sequential queries with a single constant-time operation, reducing execution time from two weeks to under five seconds and improving backfill migration performance by over 240,000x.

My full CV with a detailed project history is available on my website: <https://johnduncan.io/cv.pdf>.

Massachusetts Institute of Technology: Lincoln Laboratory (MIT: LL)

Lexington, MA

During my internship at MIT Lincoln Laboratory, I wrote Local Evacuation Alert Verification (LEAV), an Android application sponsored by the Department of Homeland Security (DHS) which assists hurricane evacuees by routing them to shelters, displaying real-time evacuation zone statuses through an interactive map, and facilitating communication between State Emergency Managers (EMs) and evacuees. Additionally, I designed an interactive module for HURREVAC-eXtended (HVX), a Federal Emergency Management Agency (FEMA)-sponsored platform, enabling EMs to dynamically update and monitor evacuation zone statuses via a web interface.

LEADERSHIP

The Cairo

Washington, DC

Outside of work, I co-led a comprehensive board transformation for The Cairo (1894), a historic building listed on the DC Inventory of Historic Sites (1990) and the National Register of Historic Places (1994), overseeing its \$1M+ annual budget. I facilitated the creation of a strategic financial management plan that included auditing finances, eliminating wasteful spending, restructuring staff, enhancing vendor accountability, and developing a proactive capital improvement strategy—all while keeping fee increases below 5%.

Additionally, I partnered with a horticulturist from Harvard's Dumbarton Oaks Gardens and a landscape architect from Oehme, van Sweden & Associates to design and install a perennial pollinator garden, incorporating heirloom plants that are concomitant with period-appropriate Victorian varieties.

Eisenhower Institute

Gettysburg, PA

During my undergraduate studies, I served as an Eisenhower Institute Undergraduate Fellow (EIUF), a program designed to cultivate political leadership skills and empower young leaders to tackle society's most challenging public policy issues.

PERSONAL PROJECTS

In my free time, I enjoy challenging myself by working on idiosyncratic projects across diverse domains. Recently, I built a Golang query builder compatible with Django ORM syntax, uncovered a bug in the Google Chromium URI parser, discovered a novel method to losslessly convert legacy Apple Macintosh data fork fonts to macOS resource forks, ported James Ashton's X-Face decoder from C to Golang, and created a Mozilla XUL/XPCOM add-on to display 'face' images from email and newsgroup message headers. I also have a passion for design and previously won an interface design competition.